

Year 10 -Pathway 2
Mini Test 5.2 -Semester 2
Statistics

Resources allowed: Calculator and 1 x A4 page one side

Name: ANSWERS Class: _____ Time: 30 minutes Total Marks: ____/26

Show working and answers on this sheet. Show working in sufficient detail to support your answers.
Incorrect answers without supporting reasoning may be allocated zero marks.

Question 1

(4 Marks)

Complete the calculations for the following measures of central tendency and measures of spread for the data set below:

7, 10, 11, 14, 17, 23

Median: $\frac{11+14}{2}$
 $= 12.5$

Q1: 10

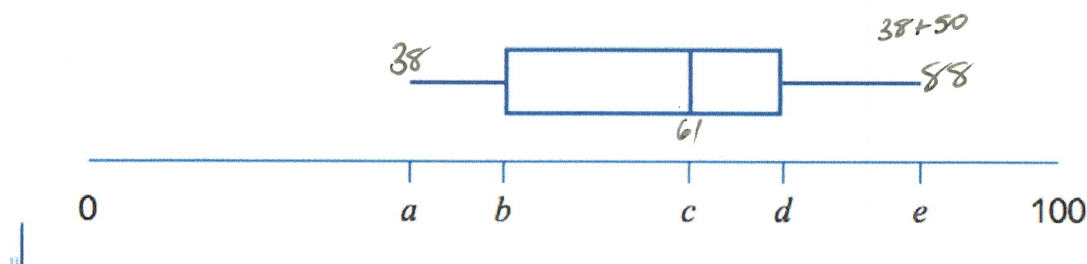
Q3: 17

Interquartile Range: $Q3 - Q1$
 $= 17 - 10$
 $= 7$

Question 2

(3, 2- 5 Marks)

The boxplot shows the distribution of test scores in a class (Class A) of 20 students.



The lowest score in the class was 38, the range of the scores was 50 and the median was 61.

- a) Write down the values of
- $a = 38$
 $c = 61$
 $e = 88$

- b) When all the test scores were added up the total was 1240. What was the mean of the test scores?

$$\frac{1240}{20} = 62$$

Question 3

(5, 5 -10 Marks)

The number of late passes issued daily by a school over a four week period were as follows:

4, 4, 4, 5, 5, 6, 6, 6, 6, 7, 7, 7, 7, 8, 8, 8, 9, 9, 10, 14

- a) Determine the five number summary for this data.

$$Q1 = \frac{5+6}{2} = 5.5$$

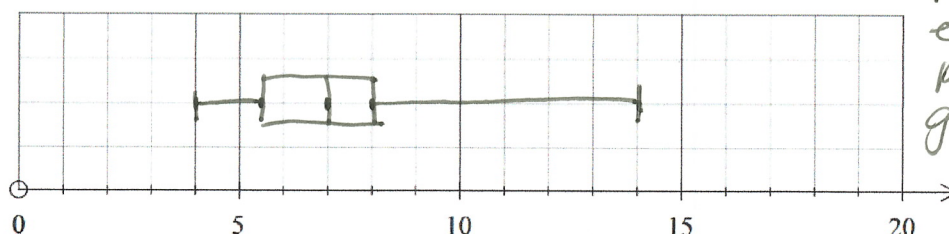
$$\text{median} = 7$$

$$Q3 = \frac{8+8}{2} = 8$$

$$\text{min} = 4$$

$$\text{max} = 14$$

- b) Use the five number summary to construct a box plot on the scale below that reflects your response in (b).

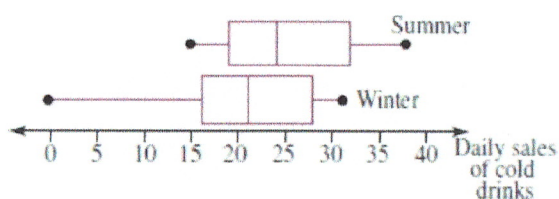


1 mark for each correct point on graph.

Question 4

(2, 2, 3 - 7 Marks)

The box-and-whisker plots below show the average daily sales of cold drinks at the school canteen in summer and winter.



- a) Calculate the range of sales in both summer and winter.

$$\text{Summer} : 38 - 15 = 23$$

$$\text{Winter} : 32 - 0 = 32$$

- b) Calculate the interquartile range of the sales in both summer and winter.

$$\text{Summer} : 33 - 19 = 14$$

$$\text{Winter} : 27 - 16 = 11$$

- c) Comment on the two data sets, both in terms of measures of centre and measures of spread.

Summer - larger IQR (middle 50% of data was more slightly more dispersed)

Winter - larger range

~ End of Test ~